

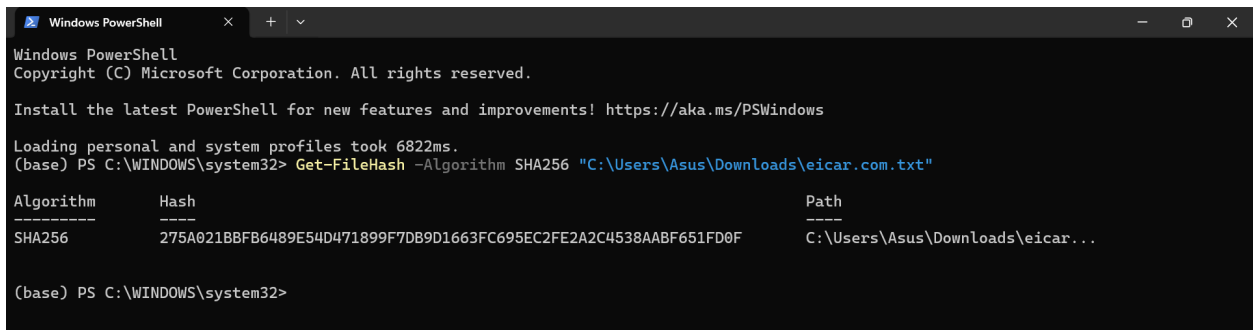
Malware Analysis

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Task: Static Malware Analysis using Free Online Tools

1. Tool Setup & File Acquisition: Download the EICAR test file from Download Anti Malware Testfile - EICAR. Save it to your Downloads folder as eicar.com.txt. Confirm your operating system and antivirus settings allow temporary access to the file

2. Hash Computation: Compute the SHA256 hash of the file using PowerShell on Windows or sha256sum on Linux/macOS. Record the output exactly as shown in your terminal.



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Loading personal and system profiles took 6822ms.
(base) PS C:\WINDOWS\system32> Get-FileHash -Algorithm SHA256 "C:\Users\Asus\Downloads\eicar.com.txt"

Algorithm      Hash
-----
SHA256         275A021BBFB6489E54D471899F7DB9D1663FC695EC2FE2A2C4538AABF651FD0F
Path
-----
C:\Users\Asus\Downloads\eicar...
```

Algorithm: SHA256

Hash: 275A021BBFB6489E54D471899F7DB9D1663FC695EC2FE2A2C4538AABF651FD0F

Path: C:\Users\Asus\Downloads\eicar...

3. Multi-Engine Scan: Submit the SHA256 hash (not the file) to VirusTotal at VirusTotal. Examine the Detection tab and document the number of engines that flag the file as malicious, along with three named detections.

We had search on VirusTotal our hash value ,then we will get these results:

Total detections (e.g., **66/68 engines detected malicious**)

3 Detection Names:

Virus/EICAR_Test_File
TestFile/Win32.EICAR
EICAR_TEST_FILE

←→↻virusotal.com/gui/file/275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f

66
/ 68
Community Score 3734

File distributed by Offensive Security

ReanalyzeSimilarMore

275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
eicar.com-23272

Size 68 B
Last Analysis Date 2 minutes ago

powerhellknown-distributoridlevia-torattachmentdetect-debug-environmentdirect-cpu-clock-accesslong-sleeps

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY30+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Code insights

EICAR is a test string used to detect and test antivirus software. It's like a "dummy virus" that triggers an antivirus engine to react, demonstrating its ability to identify and neutralize threats.
Here's the key:
It's NOT a real virus. EICAR is harmless and cannot infect your computer.
It's a standardized test: Almost all antivirus programs are designed to recognize EICAR as a potential threat, ensuring they're working properly.
Show more

Popular threat labelvirus.eicar.testThreat categoriesvirustrojanFamily labelseicar.test.file

Security vendors' analysisDo you want to automate checks?

AhnLab-V3Virus/EICAR_Test_FileAlibabaVirus/Win32/EICAR.A

AhnLab-V3	Virus/EICAR_Test_File	Alibaba	Virus/Win32/EICAR.A
AliCloud	Engtest:Multi/Eicar	ALYac	Misc.Eicar-Test-File
Antiy-AVL	TestFile/Win32.EICAR	Arcabit	EICAR-Test-File (not A Virus)
Avast	EICAR Test- NOT Virus!!!	Avast-Mobile	Eicar
AVG	EICAR Test- NOT Virus!!!	Avira (no cloud)	Eicar-Test-Signature
BitDefender	EICAR-Test-File (not A Virus)	Bkav Pro	W32.EicarTest.Trojan
ClamAV	Eicar-Test-Signature	CMC	Eicar.test.file
CTX	Txt.virus.eicar	Cynet	Malicious (score: 99)
DrWeb	EICAR Test File (NOT A Virus!)	Elastic	Eicar
Emsisoft	EICAR-Test-File (A)	eScan	EICAR-Test-File
ESET-NOD32	Eicar Test File	Fortinet	EICAR_TEST_FILE
GData	EICAR_TEST_FILE	Google	Detected
Gridinsoft (no cloud)	Trojan.U.EICAR_Test_File.dd	Huorong	TEST/AVEngTestFile/EICAR
Ikarus	EICAR-Test-File	Jiangmin	EICAR-Test-File

Basic properties

MD544d88612fea8a8f36de82e1278abb02f

SHA-13395856ce81f2b7382dee72602798b642f14140

SHA-256275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f

SSDEEP3:a:JraNvgsgVq5wHq9:3JuOgrsko

TLSH7141A022003B0EEF2BA20B00200032E8B00808020E2CE0A3820A020B8C83308803EC228

File typePowerShellsourcepowerhellpspsl

MagicEICAR virus test files

TrIDEICAR antivirus test file (100%)

MagikaPOWERSHELL

File size68 B (68 bytes)

History

First Seen In The Wild2012-06-14 17:18:14 UTC

First Submission2006-05-22 12:42:02 UTC

Last Submission2026-05-01 04:31:33 UTC

Last Analysis2026-05-01 04:31:33 UTC

Names

eicar.com-23272

eicar.com-18948

eicar.com-13183

eicar.com

eicar.com-8504

eicar.com-2916

TieringEngineService.exe

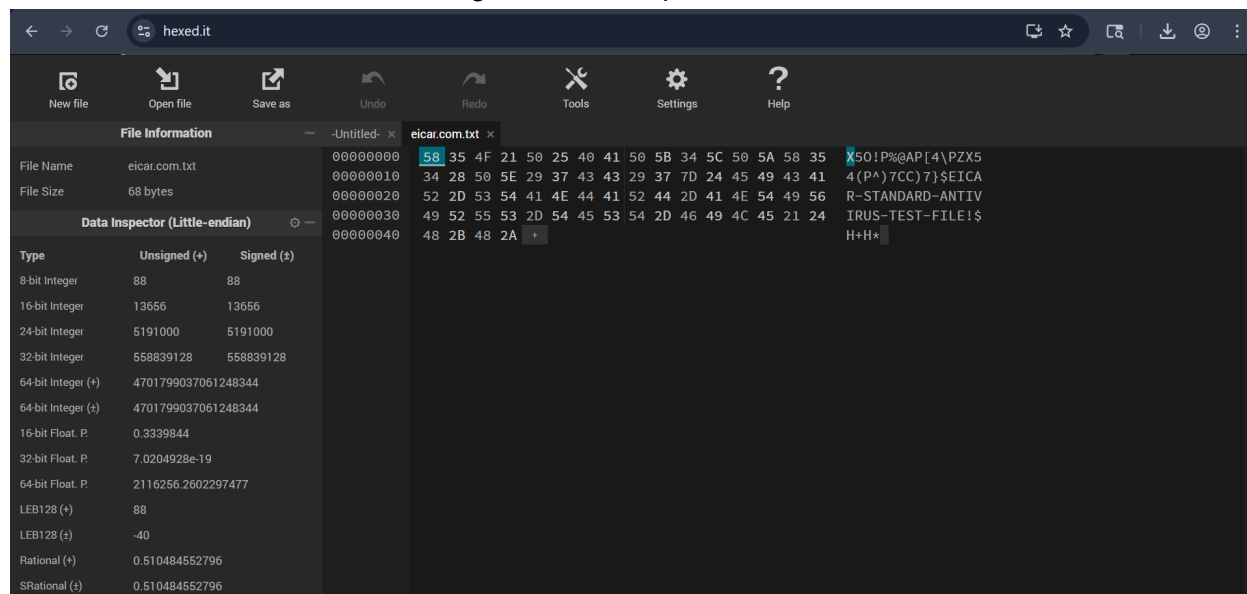
eicar.com-47193

eicar.com-42913

eicar.com-38451

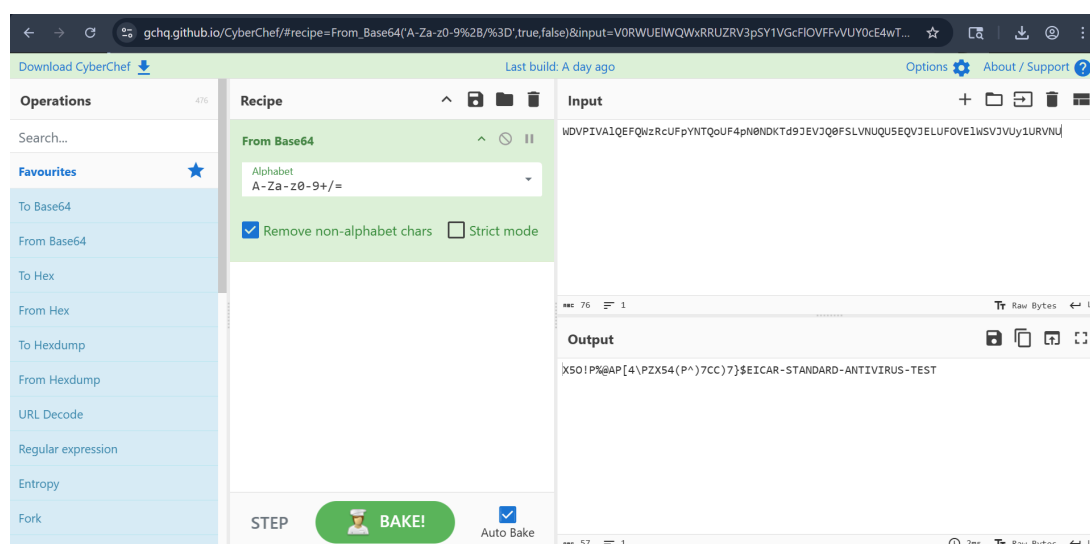
4. String & Hex Inspection: Open the file in Hexed.it (HexEd.it) and observe both the raw hexadecimal bytes and the ASCII representation. Identify and record all readable strings.

Now we had search [hexEd.it](https://hexed.it), and we had upload the [eicar.com](https://www.eicar.com).txt ,and we had observed that On left we have Hex value and on right we have equivalent ASCII text.



5. Decoding Practice: Use CyberChef (CyberChef) with the From Base64 operation to decode the following sample string and observe the output:

WDVPIVA1QEFQWzRcUFpYNTQoUF4pN0NDKTd9JEVJQ0FSLVNUQU5EQVJELUFOVEIWSVJV Uy1URVNU



Output will be:

X50!P%AP[4\ZX54(P^7CC7)\$EICAR-STANDARD-ANTIVIRUS-TEST

Indian Cyber Law and Compliance

Company Liability: Under Section 43A of the IT Act, 2000, can Helios Financial Services be legally penalised for failing to protect customer financial information that was leaked as a result of the malware infection? Explain the conditions under which liability is established.

In the event of a data breach involving customer information caused by a malware attack, Helios Financial Services could be held accountable under Section 43A of the IT Act, 2000.

This is the case when the company was processing sensitive personal data and did not have adequate security measures in place to protect it. If the data breach occurred due to negligence, the company could be ordered to pay damages to the affected customers.

Mandatory Reporting: According to the CERT-In Directions of April 2022, what are the exact steps and the maximum permitted timeframe within which Helios must report this cyber incident to the government authorities?

Under the guidelines of CERT-In, the company is obligated to report cyber incidents to the government.

The report has to be submitted within 6 hours of the detection of the incident.

Steps are:

- The incident identified

- Gathering pertinent logs and evidence

- Providing information such as type of attack, systems affected, actions taken.

This prompt reporting enables authorities to respond more quickly and contain the impact of cyber attacks.

Questions

Answer the following five questions:

1. Why is computing a cryptographic hash (such as SHA256) one of the very first steps in static malware analysis? What are the limitations of relying on hash-based detection alone, and how can attackers easily bypass it?

Ans- One of the first steps in malware analysis is to compute a cryptographic hash such as SHA256, which provides a unique identity for the file much like a fingerprint. The hash values

will tell you if two files that appear to be the same are in fact the same or not. Analysts can then query databases such as VirusTotal using this hash to quickly assess whether the file has been previously identified as malicious.

However, detection based on hashes has some limitations. If even one character in the file is changed, the hash value changes entirely. This means that attackers can easily circumvent detection by slightly changing the malware, e.g. through obfuscation or repacking. Therefore, hash matching, by itself, is not sufficient and analysts have to use other techniques too.

2. Differentiate between static and dynamic malware analysis. Give one realistic scenario where static analysis alone would be insufficient and explain what dynamic analysis would reveal in that situation.

Ans- Static analysis is when you look at a file without executing it. Dynamic analysis is when you run the file in a safe environment and observe what it does. Static analysis is fast and safe, and can reveal fundamental information like the file structure, strings, and metadata. But it does not show how the malware behaves when it is actually run. On the other hand dynamic analysis enables the analyst to see what the malware actually does for example creating files, changing the system or connecting to the internet.

For instance, if a malware file is packed or encrypted, static analysis may not disclose anything useful. In that case, its real behavior can be exposed through dynamic analysis, for example contacting a command and control server or stealing data when it is executed.

3. What are Indicators of Compromise (IOCs) and why are they central to a SOC analyst's workflow? Give at least four distinct types of IOCs and provide one example of each.

Ans- Indicators of Compromise (IOCs) are signs that a system may have been attacked or infected with malware. These are very important for SOC analysts, helping to detect threats and prevent further damage.

The most common types of IOCs are

File Hash: An unique value which identifies a file Example: 275A021BBFB648...

IP Address - The address used by malware to communicate

192.168.1.100

Domain Name: Malicious websites attacking

bad-site.com

File path: Where the malware is stored

Example: C:\Users\AppData\maleware.exe

These IOCs are shared between systems to help with the faster detection of similar attacks in the future.

4. Explain the concepts of obfuscation and packing in the context of malware. Why do attackers use these techniques, and what tools or signs would help an analyst detect that a binary has been packed?

Ans-Attackers use obfuscation and packing to hide malware and make analysis more difficult. Obfuscation means to make the code confusing or unreadable so it is hard to understand. Packing is when the malware is compressed or encrypted, the actual content is not revealed until it is run. Attackers use these techniques to avoid being caught by antivirus software and to slow down analysts.

An analyst can detect packed malware using tools such as Detect It Easy or PEStudio. Some common signs are high entropy (random looking data) , very few readable strings , or an unusual file structure .

5. Imagine you are the SOC analyst for Helios Financial Services and your static analysis has confirmed that the suspicious attachment is malicious. Outline the immediate next three steps you would take, in order, and justify why each step is important.

Ans- If I was a SOC analyst at Helios Financial Services this is what I would do:

First I would isolate the infected system from the network to contain the threat. This stops the malware from spreading to other systems.

Secondly, I would extract and block IOCs such as file hashes, IP addresses and domains. These would be added to security tools such as firewalls and anti-virus systems to prevent further attacks.

Third, I would report the incident and start an investigation. This includes alerting the security team, logging the attack and investigating how the malware gained entry to the system so that similar attacks can be prevented from happening in the future.